

Project Dataset DSA 201 Adam Faulk

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```
# install.packages("baseballr")
library(baseballr)
library(dplyr)
# install.packages("tidyverse")
library(tidyverse)
library(ggplot2)
library(lubridate)
library(janitor)
library(knitr)
```

```
data <- bref_daily_batter("2023-03-30", "2023-10-01")
data %>%
  dplyr::glimpse()
```

```
## Rows: 660
## Columns: 30
## $ bbref_id <chr> "sema01", "acuna01", "freemf01", "bregma01", "lowena01~
## $ season <int> 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2~
## $ Name <chr> "Marcus Semien", "Ronald Acuña Jr.", "Freddie Freeman", "Alex~
## $ Age <dbl> 32, 25, 33, 29, 27, 29, 30, 25, 26, 22, 24, 23, 31, 30, 30, 2~
## $ Level <chr> "Maj-AL", "Maj-NL", "Maj-NL", "Maj-AL", "Maj-AL", "Maj-NL", "~
## $ Team <chr> "Texas", "Atlanta", "Los Angeles", "Houston", "Texas", "Atlan~
## $ G <dbl> 162, 159, 161, 161, 161, 162, 160, 158, 159, 155, 162, 158, 1~
## $ PA <dbl> 753, 735, 730, 724, 724, 720, 720, 718, 715, 714, 708, 694, 6~
## $ AB <dbl> 670, 643, 637, 622, 623, 608, 585, 638, 636, 654, 568, 641, 5~
## $ R <dbl> 122, 149, 131, 103, 89, 127, 108, 93, 117, 102, 97, 97, 68, 1~
## $ H <dbl> 185, 217, 211, 163, 163, 172, 115, 171, 179, 180, 156, 177, 1~
## $ X1B <dbl> 112, 137, 121, 106, 105, 88, 48, 123, 107, 109, 88, 108, 88, ~
## $ X2B <dbl> 40, 35, 59, 28, 38, 27, 19, 36, 32, 37, 32, 28, 29, 40, 36, 3~
## $ X3B <dbl> 4, 4, 2, 4, 3, 3, 1, 7, 3, 2, 1, 11, 0, 1, 5, 4, 5, 4, 0, 1, ~
## $ HR <dbl> 29, 41, 29, 25, 17, 54, 47, 5, 37, 32, 35, 30, 22, 39, 24, 21~
## $ RBI <dbl> 100, 106, 102, 98, 82, 139, 104, 54, 97, 103, 109, 96, 96, 10~
## $ BB <dbl> 72, 80, 72, 92, 93, 104, 126, 70, 59, 47, 132, 40, 70, 96, 73~
## $ IBB <dbl> 1, 3, 12, 1, 2, 8, 5, 2, 4, 6, 11, 1, 0, 3, 22, 2, 2, 0, 6, 6~
## $ uBB <dbl> 71, 77, 60, 91, 91, 96, 121, 68, 55, 41, 121, 39, 70, 93, 51, ~
## $ SO <dbl> 110, 84, 121, 87, 165, 167, 215, 75, 172, 175, 129, 121, 214, ~
## $ HBP <dbl> 5, 9, 16, 8, 5, 4, 6, 3, 9, 11, 2, 4, 15, 8, 1, 6, 6, 14, 3, ~
## $ SH <dbl> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0~
## $ SF <dbl> 5, 3, 5, 2, 3, 4, 3, 6, 11, 2, 5, 8, 11, 5, 6, 6, 1, 5, 3, 5, ~
## $ GDP <dbl> 5, 15, 14, 22, 22, 13, 4, 9, 18, 14, 18, 11, 10, 5, 8, 12, 12~
## $ SB <dbl> 14, 72, 23, 3, 1, 1, 0, 21, 3, 37, 12, 49, 2, 14, 28, 14, 30, ~
```

```
## $ CS      <dbl> 3, 14, 1, 1, 0, 0, 2, 3, 1, 10, 5, 15, 1, 3, 6, 3, 0, 7, 2, 2~
## $ BA      <dbl> 0.276, 0.337, 0.331, 0.262, 0.262, 0.283, 0.197, 0.268, 0.281~
## $ OBP     <dbl> 0.348, 0.416, 0.410, 0.363, 0.360, 0.389, 0.343, 0.340, 0.345~
## $ SLG     <dbl> 0.478, 0.596, 0.567, 0.441, 0.414, 0.604, 0.474, 0.370, 0.516~
## $ OPS     <dbl> 0.826, 1.012, 0.976, 0.804, 0.775, 0.993, 0.817, 0.710, 0.861~
```

```
dim(data)
```

```
## [1] 660 30
```

```
## Sort by batting average to see who batted the highest
highest_ba <- subset(data, G >= 81)
highest_ba <- highest_ba[order(-highest_ba$BA), ]
```

This is just a little bit of EDA to understand the data before I go into more detail with my code. For some background on what my final project is hoping to accomplish. I am working with the Baltimore Orioles in the 2023 season to see what players the Orioles should target at the deadline. In 2023 the Orioles were surging, and had a good chance of making the playoffs at the deadline. This meant that they were buyers at the deadline, looking for some players they could bring in to fill in the gaps for the team. My goal is to indentify the gaps and then identify potential players that could fill these gaps.

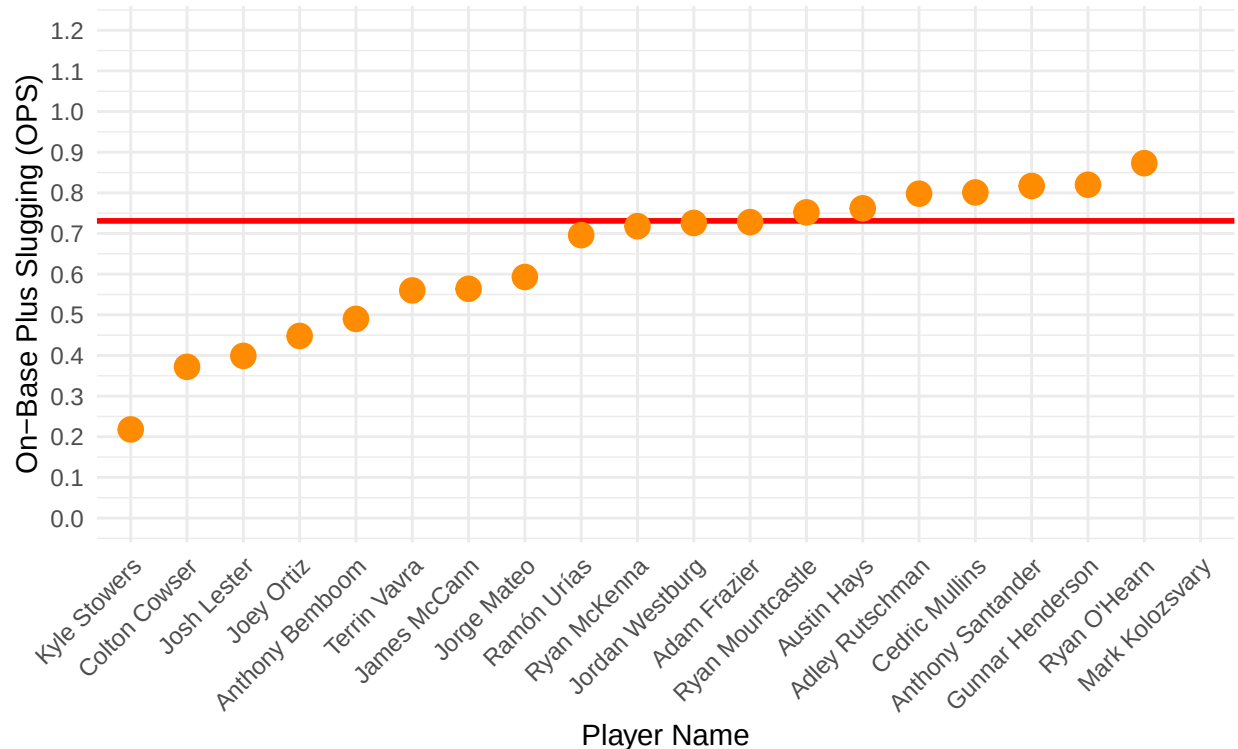
```
# Get the data about stats before the trade deadline and then look at just the orioles.
deadlinestats <- bref_daily_batter("2023-03-30", "2023-08-01")
orioles <- deadlinestats %>% filter(Team == "Baltimore")
```

Now that we have data for just the orioles, let's look at all of their players compared to the league average OPS. With a quick google search, I found that at the trade deadline in 2023, the league average OPS was .731. I will add a red line in my graph to represent this.

```
# Create the scatter plot
ggplot(orioles, aes(x = reorder(Name, OPS), y = OPS)) +
# Add a red line to represent league average OPS
geom_hline(yintercept = 0.731, color = "red", linetype = "solid", linewidth = 1) +
# Add the points (Orioles orange)
geom_point(size = 4, color = "darkorange") +
# Customizing the axes
labs(
  title = "2023 Orioles: OPS by Player",
  subtitle = "Red line represents 2023 league average OPS",
  x = "Player Name",
  y = "On-Base Plus Slugging (OPS)"
) +
# Set a logical limit for the Y axis
scale_y_continuous(limits = c(0, 1.2), breaks = seq(0, 1.2, by = 0.1)) +
theme_minimal() +
# Rotate the x-axis labels so they are readable
theme(
  axis.text.x = element_text(angle = 45, hjust = 1)
)
```

2023 Orioles: OPS by Player

Red line represents 2023 league average OPS



With this graph, we can look at what the orioles did good and where they need to improve. Unfortunately my dataset does not include whether the batters are left or right handed batters, so I will do some research online and then create a left handed orioles dataset and a right handed orioles dataset. Switch hitters will be included in their own dataset to remove any bias.

```
righthandedorioles <- orioles %>% filter(Name %in% c("Austin Hays", "Ryan Mountcastle", "Jorge Mateo", "
lefthandedorioles <- orioles %>% filter(Name %in% c("Gunnar Henderson", "Cedric Mullins", "Adam Frazier
switchorioles <- orioles %>% filter(Name %in% c("Adley Rutschman", "Anthony Santander"))

lefty_total_ops <- lefthandedorioles %>%
  ungroup() %>%
  summarise(
    group_obp = sum(H + BB + HBP) / sum(AB + BB + HBP + SF),
    group_slg = sum(H + X2B + 2 * X3B + 3 * HR) / sum(AB),
    group_ops = group_obp + group_slg
  )
righty_total_ops <- righthandedorioles %>%
  ungroup() %>%
  summarise(
    group_obp = sum(H + BB + HBP) / sum(AB + BB + HBP + SF),
    group_slg = sum(H + X2B + 2 * X3B + 3 * HR) / sum(AB),
    group_ops = group_obp + group_slg
  )

switch_total_ops <- switchorioles %>%
```

```

ungroup() %>%
summarise(
  group_obp = sum(H + BB + HBP) / sum(AB + BB + HBP + SF),
  group_slg = sum(H + X2B + 2 * X3B + 3 * HR) / sum(AB),
  group_ops = group_obp + group_slg
)

```

We now have the orioles into three different groups, and this allows us to now view the stats of each type of hitter. This is important because in a world of matchups, knowing if you need to improve against left handed pitching or right handed pitching is crucial. Left handed batters are going to be better against right handed pitchers, and right handed batters are going to be better against left handed pitching. Let's go into more detail with code

```

#OPS for all left handed batters on the orioles
lefty_total_ops <- lefthandedorioles %>%
  ungroup() %>%
  summarise(
    group_obp = sum(H + BB + HBP) / sum(AB + BB + HBP + SF),
    group_slg = sum(H + X2B + 2 * X3B + 3 * HR) / sum(AB),
    group_ops = group_obp + group_slg
  )
#OPS for all right handed batters on the Orioles
righty_total_ops <- righthandedorioles %>%
  ungroup() %>%
  summarise(
    group_obp = sum(H + BB + HBP) / sum(AB + BB + HBP + SF),
    group_slg = sum(H + X2B + 2 * X3B + 3 * HR) / sum(AB),
    group_ops = group_obp + group_slg
  )
#OPS for all switch hitters on the Orioles
switch_total_ops <- switchorioles %>%
  ungroup() %>%
  summarise(
    group_obp = sum(H + BB + HBP) / sum(AB + BB + HBP + SF),
    group_slg = sum(H + X2B + 2 * X3B + 3 * HR) / sum(AB),
    group_ops = group_obp + group_slg
  )
#Adding into one dataset so it is easier to compare
orioles_ops_comparison <- bind_rows(
  lefty_total_ops %>% mutate(Batter_Side = "Left"),
  righty_total_ops %>% mutate(Batter_Side = "Right"),
  switch_total_ops %>% mutate(Batter_Side = "Switch")) %>%
  select(Batter_Side, everything())
# 2. Convert to 'factors' to force a specific order on the X-axis (optional, but cleaner)
orioles_ops_comparison$Batter_Side <- factor(orioles_ops_comparison$Batter_Side,
  levels = c("Left", "Switch", "Right"))

# Creating a ggplot visualization to make the final graph cleaner. Originally used the function "Kable"

ggplot(orioles_ops_comparison, aes(x = Batter_Side, y = group_ops, fill = Batter_Side)) +
  geom_col(color = "black", show.legend = FALSE) + # geom_col makes the bars based on the values
  geom_text(aes(label = round(group_ops, 3)), # Add numeric labels on top of the bars
    vjust = -0.5, # Position labels slightly above the bar

```

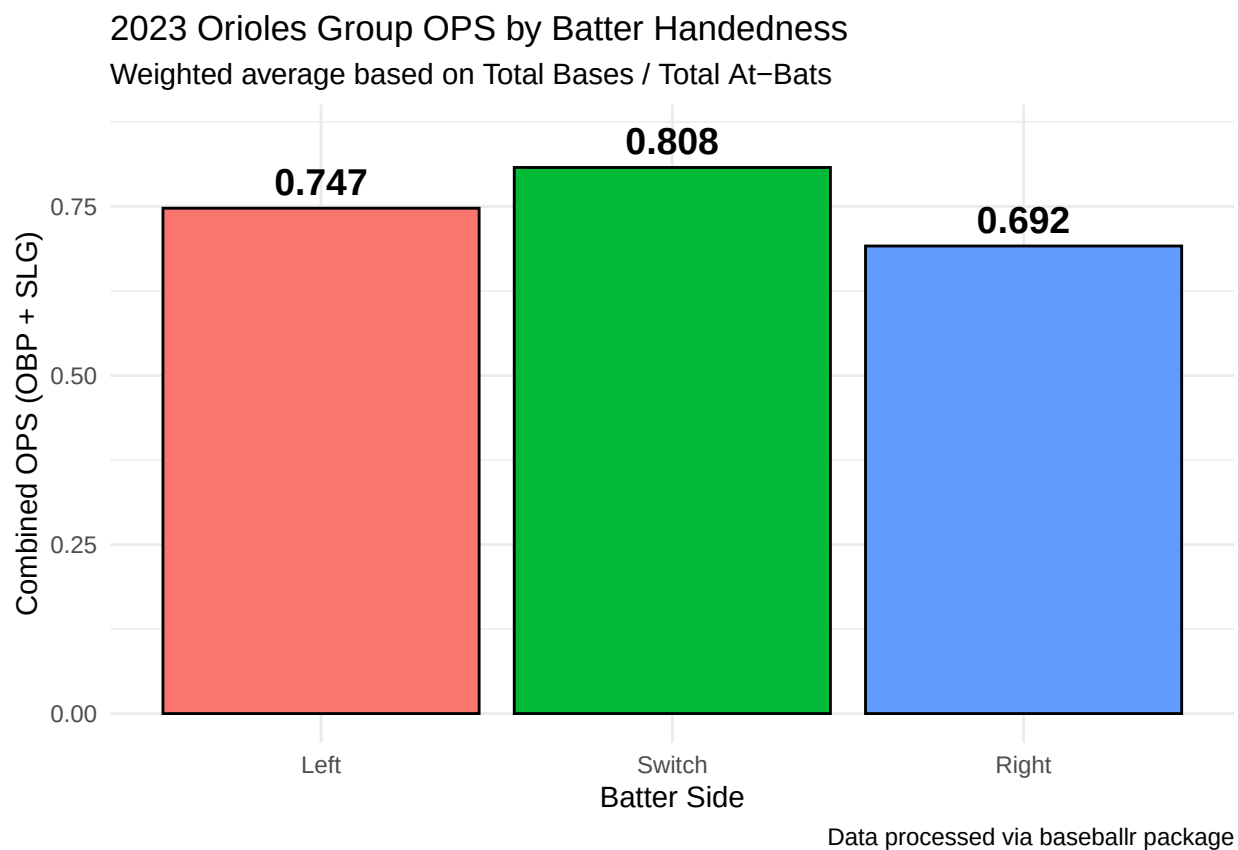
```

size = 5, # Adjust label size
fontface = "bold") +

# Styling and Titles
theme_minimal() + # Uses a clean white background
labs(title = "2023 Orioles Group OPS by Batter Handedness",
      subtitle = "Weighted average based on Total Bases / Total At-Bats",
      x = "Batter Side",
      y = "Combined OPS (OBP + SLG)",
      caption = "Data processed via baseballr package") +

# Adjust Y-Axis so there is room for the labels
expand_limits(y = c(0, max(orioles_ops_comparison$group_ops) + 0.05))

```



From these results we can see that right handed batters are the area that needs the most improvement on the orioles. From a quick google search, the average OPS from the 2023 season was .731, so .692 is well below that. We now know that we need to target right-handed batters that have an OPS above .731. We want to target players that are above .731 because we want above average players who can improve our chances to make the World Series. The code below will have `#` to give more detail, but I will narrow down to some players that we should target

```

potential_targets <- deadlinestats %>%
  filter(OPS > 0.731)
# This turned back 215 observations, with many players having played only a handful of games. Since thi
potential_targets <- deadlinestats %>%
  filter(OPS > 0.731) %>%

```

```

filter(G >= 50)
# I now am going to run all of these player names through gemini to find who was on an expiring contract

# 1. Define the list of players on expiring deals in 2023
expiring_players <- c(
  "Shohei Ohtani", "Matt Chapman", "Cody Bellinger", "J.D. Martinez",
  "Jorge Soler", "Andrew McCutchen", "Brandon Belt", "Justin Turner",
  "Tommy Pham", "Charlie Blackmon", "C.J. Cron", "Gary Sánchez",
  "Travis Jankowski", "Donovan Solano", "Randal Grichuk", "Adam Duvall",
  "Eddie Rosario", "Kevin Kiermaier", "Joey Votto"
)

# 2. Filter data so only these names remain
expiring_stats <- deadlinestats %>%
  filter(Name %in% expiring_players)

# 3. View the new dataset
view(expiring_stats)

```

We are now down to 19 potential targets. This is good but we are going to have to narrow it down a little more. Some of these teams are not going to want to sell at the deadline. Let's narrow it down to teams that are below .500 record wise.

```

expiring_stats <- deadlinestats %>%
  filter(Name %in% expiring_players) %>%
  filter(Team %in% c("Colorado", "Pittsburgh", "New York"))
view(expiring_stats)

#Creating a ggplot visualization to show which players to showcase
library(ggplot2)

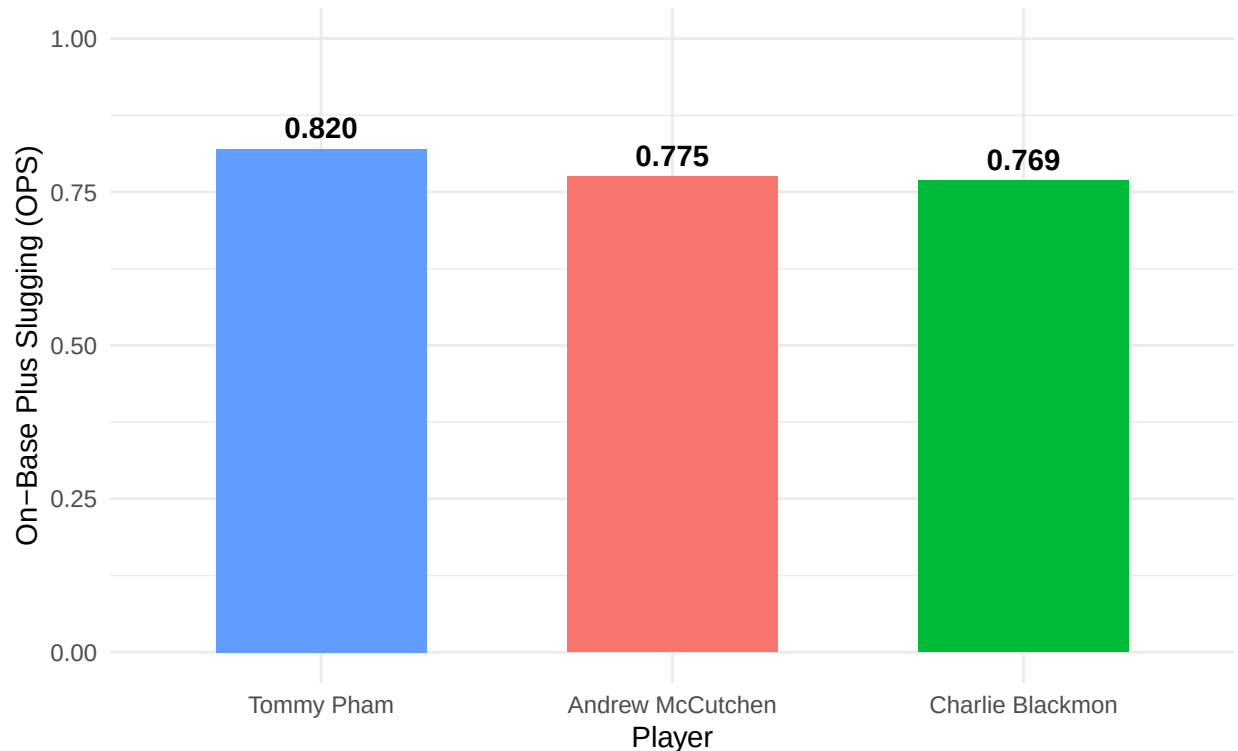
# 1. Create a subset of just these three players
expiring_stats <- deadlinestats %>%
  filter(Name %in% c("Andrew McCutchen", "Tommy Pham", "Charlie Blackmon"))

# 2. Build the plot
ggplot(expiring_stats, aes(x = reorder(Name, -OPS), y = OPS, fill = Name)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  # Add the actual OPS values on top of the bars
  geom_text(aes(label = sprintf("%.3f", OPS)), vjust = -0.5, fontface = "bold") +
  # Customizing the look
  labs(
    title = "Target Acquisitions: OPS Comparison",
    subtitle = "Expiring contracts on selling teams at the 2023 Deadline",
    x = "Player",
    y = "On-Base Plus Slugging (OPS)"
  ) +
  theme_minimal() +
  # Setting the y-axis slightly higher than the max OPS for breathing room
  scale_y_continuous(limits = c(0, 1.0))

```

Target Acquisitions: OPS Comparison

Expiring contracts on selling teams at the 2023 Deadline



We are now down to 3 players. Andrew McCutchen, Tommy Pham, and Charlie Blackmon. Players who had multiple teams listed were omitted because this meant that they had already been traded during the season, making a second trade unlikely. The conclusion from this is that as Tommy Pham and Andrew McCutchen are both right handed we should target both of them. We can also look at acquiring Charlie Blackmon, but since he is left handed this is less of a priority. This addresses our stakeholders need (The Baltimore Orioles) of improving the roster at the trade deadline so that we can make a playoff push.

This raises some more questions like what are the other teams in the AL East's average OPS? Are they better left handed or right handed. What about our matchups towards the end of the season. If we are facing teams down the stretch that play more right handed pitchers then we should also target a lefty. This is just the tip of the iceberg, but this data also answers a very important question that we started with. We need to target a right handed batter to improve our team OPS from the right side of the plate. There are not many ethical questions to worry about here, if any. Since we are using statistics, this takes away the bias that people may have and gives a more objective approach. There could potentially be issues with me only looking at players who played at least 50 games, as injuries are out of players' control, but I felt this was fair to do as small sample sizes can skew the data.

I chose to work with this dataset as I am a huge sports fan, and I would love to work in sports as my full time job. My dream is to work as a data analyst for a professional sports team, ideally the orioles.